

Hot Days Turn Deadlier Where Guns Are Easy to Carry

Lessons from “Access to Guns in the Heat of the Moment: More Restrictive Gun Laws Mitigate the Effect of Temperature on Violence” by Jonathan Colmer & Jennifer L. Doleac in *The Review of Economics and Statistics*, 2026.

Hot days lead to more gun homicides, but stricter concealed-carry laws cut these increases from 6% to 2%. On an average day, one degree of extra heat means more homicides nationwide — except in states that restrict who can carry a concealed gun, where that heat-driven increase lessens.

BACKGROUND

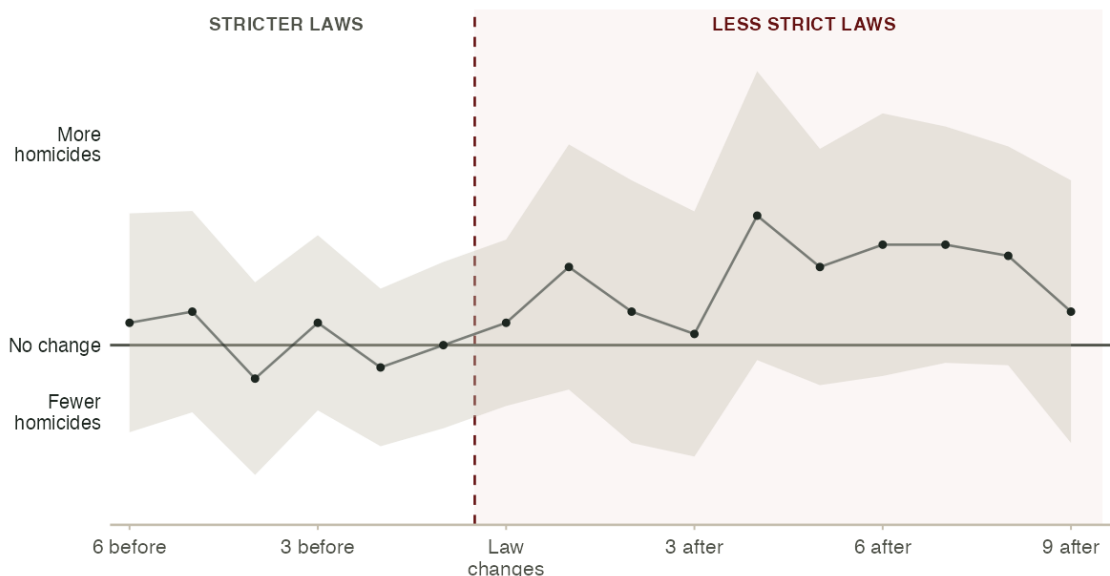
Past research has established that high temperatures lead to increased aggression and violent behavior: hotter days are linked to more confrontations and violent crime. Additionally, states differ significantly in their **concealed-carry laws** — the rules governing who can carry a hidden handgun in public. Some states let nearly any adult carry one; others require a permit that officials can deny, or ban concealed carry altogether. Together, these two facts raise an important question: if heat makes people more likely to become violent, does having a gun close at hand change the results of these confrontations? If so, then stricter concealed carry laws may be especially beneficial as areas get hotter.

To answer this question, researchers matched 25 years of daily police crime reports (1991–2016) with local weather records and each state’s concealed-carry laws, month by month. This allowed them to see the temperature, time, and laws in place for each crime that was committed. They focused on homicide because it is the crime type most consistently and reliably reported to police, across both place and time.

WHAT THE STUDY FOUND

HOW MUCH DEADLIER HOT DAYS BECOME AFTER A STATE LOOSENS ITS CONCEALED-CARRY LAW

Source: Colmer & Doleac (2026).



Environmental Inequality Lab · Access to Guns in the Heat of the Moment, 2026

Note: Each point tracks the gap in hot-day homicides between states that eventually loosen their concealed-carry law and states that keep theirs strict, year by year before and after the change (year 0). Before the change, the gap is flat — hot days are equally dangerous in both groups. Right after, it opens and stays open for years, which is what makes this look like the law’s effect rather than a coincidence.

It is difficult to measure how a law might affect heat-induced gun violence because it is impossible to measure the difference

between an area if the law had never been passed because there are many possibilities as to why the law might have been passed at that exact time. It would also be difficult to compare states with different laws and/or temperatures for the same reason: there might be other circumstances (e.g. cultural, political, poverty) that would lead to different outcomes that could also affect violence.

However, researchers found a solution that allowed them to judge the actual effects they wanted to observe. They compared how much *extra* violence a hot day brings in strict-law places versus loose-law places. That comparison cancels out whatever else makes those states different, as long as those other differences don't also change how a place responds to heat specifically. As a check, they also tracked the same states over time, comparing each state's own hot-day homicides before and after it changed its own law — shown in the figure above. Both approaches point to the same conclusion.

Nationally, each one-degree-Celsius increase in daily temperature is tied to roughly six percent more homicides than usual on an average day. In states with stricter concealed-carry laws, that heat-driven bump nearly disappears — both when comparing different states at the same time, and when tracking the same states before and after they changed their own laws. **Stricter concealed-carry laws blunt most of the effect of heat on homicides.**

This pattern is also concentrated specifically in gun homicides — not knives or other weapons, and not aggravated assaults broadly. That suggests strict laws aren't stopping hot-day arguments from happening in the first place; they're making the arguments that do happen less likely to turn deadly.

TAKEAWAYS

A GROWING RISK

Concealed-carry laws have loosened nationwide since the 1990s — from 36 states with strict laws in 1991 to just 10 by 2016 but climate change is bringing more hot days every year. If the whole country still had 1991's stricter mix of laws today, a one-degree hot day would cause an estimated 97 additional homicides nationwide; with today's looser laws in place, that same hot day causes roughly 299. **Fewer states now have the laws that lessen heat's effect on violence, just as rising temperatures make that protection more important.**

PUTTING A PRICE ON A HOT DAY

Using standard estimates of the cost of a homicide to society (about \$11.3 million each), the authors calculate that avoiding these heat-driven deaths nationwide would be worth roughly \$3.4 billion for every one-degree-Celsius rise in temperature — comparable to the estimated social benefit of hiring nearly 13,000 more police officers. The authors are clear this is a rough, illustrative calculation, not a precise cost-benefit accounting. **Even by conservative estimates, the stakes of heat-driven gun violence are large.**

AGGRESSION, NOT MORE ENCOUNTERS

Why does heat make hot days more lethal where guns are easy to carry? It could be that people simply see more people when the weather is nice, or that heat makes people more aggressive once a conflict starts. The evidence favors the latter. The effect holds for homicides at home when they aren't necessarily seeing more people, and the effect is strongest in the hottest hours of the day, which doesn't point to an increase in social interactions as strongly. **The heat makes people more aggressive, which turns an ordinary conflict into lethal one when guns are nearby.**

WHAT THIS DOES (AND DOESN'T) SHOW

This study is about impulsive, heat-driven homicides. It doesn't speak to premeditated crime or gun violence more broadly. It also doesn't weigh the costs to gun owners of more restrictive laws, or say whether stricter concealed-carry laws should be adopted; that tradeoff is a policy judgment the authors leave open. **What the research does show is that temperature is a real, measurable driver of gun violence — and how much it matters depends on the laws already in place.**